

EVIDENCE - BASED COACHING MANUAL

THE LEAN MACHINE PROTOCOL

A 10-Week Body Recomposition System — Built for You

81→75

KG TARGET

~22→15

% BODY FAT

10

WEEKS

6-Day

PPL / LPP

Profile: 28 y • 174 cm • Advanced • Vegetarian • AM Training • 10k steps/day

1 The Training Split

28 yrs
Age

174 cm
Height

81 kg
Weight

~22%
Body fat

3+ yrs
Advanced

Vegetarian
+ dairy/whey

7–8 h
Sleep

10,000
Steps/day

YOUR SPLIT:

6-Day PUSH / PULL / LEGS (Legs-first / LPP option included)

You chose PPL/LPP — an excellent, time-tested hypertrophy structure. Run twice through the week it delivers the key **2×/week per-muscle frequency** that supports muscle retention in a deficit. The trade-off vs. Upper/Lower is higher systemic fatigue across 6 days, so fatigue management (below) is non-negotiable while dieting.

How PPL/LPP Stacks Up

Criterion	Your 6-Day PPL/LPP	Why it matters
Per-muscle frequency	2×/week	Schoenfeld 2016 meta: 2× beats 1× when volume equated — key for retaining muscle
MPS coverage	Every 3–4 days	Keeps muscle protein synthesis elevated; avoids the 6-day "off" gap of bro-splits
Volume per muscle	High, well-distributed	Split across 2 sessions = quality reps, less per-session junk volume
Muscle focus / specialization	Excellent	Dedicated push, pull & leg days allow targeted weak-point work
Strength retention (deficit)	Good (keep heavy "A" days)	Heavy compounds early in each session preserve strength & muscle
Fatigue management	Needs discipline	6 days + 10k steps + deficit = high load — use the fallback if under-recovered
Sustainability	High if recovery is managed	Enjoyable, flexible order (PPL or LPP)

About the "muscle loss" you noticed — likely NOT real muscle

Early in a deficit, looking flatter/smaller is almost always **glycogen + water depletion** (each gram of glycogen holds ~3 g water), not lost muscle. It reverses within 24–48 h of eating carbs. **Judge by strength, not the mirror:** if your working loads & reps hold week-to-week, your muscle is preserved. Even genuine losses return fast via muscle memory (retained myonuclei). Keep most carbs around training to stay full.

Bottom line

PPL/LPP gives you 2× frequency, strong volume distribution, and dedicated focus days. Keep the heavy compounds heavy, manage fatigue ruthlessly in the deficit, and it will drive recomposition just as well as Upper/Lower — with the style you prefer.

2 Weekly Layout

MON
Push A

TUE
Pull A

WED
Legs A

THU
Push B

FRI
Pull B

SAT
Legs B

SUN
Rest

Day	PPL order	LPP order (your preference)	Cardio / Steps
Mon	Push A	Legs A	10k steps PM
Tue	Pull A	Push A	10k steps PM
Wed	Legs A	Pull A	10k + optional 20-30 min LISS
Thu	Push B	Legs B	10k steps PM
Fri	Pull B	Push B	10k steps PM
Sat	Legs B	Pull B	10k + optional LISS
Sun	REST	REST	10k steps (easy)

Same 6 sessions — just pick your starting point. "A" days lead with heavier compounds (strength-leaning); "B" days emphasize a complementary angle + more volume.

Fatigue fallback (use it in the deficit)

6 training days + 10k steps on ~600 kcal deficit is high systemic load. If strength drops, sleep worsens, or joints ache, switch to a rolling **Push → Pull → Legs → REST → repeat** (1 rest every 3 sessions) or drop to 5 days. Do not grind 6 days while under-recovered.

Timeline reconciliation

Diet/deficit runs the full **10 weeks**. Training is an **8-week mesocycle** (deload at Week 5), then Weeks 9–10 are a short peak/finish block.

3 The Workouts

Legend: RPE on Reps-in-Reserve scale (RPE 8 = 2 reps left). Tempo = eccentric-pause-concentric. Sessions capped at 7 exercises to manage fatigue in the deficit.

Push A — Chest Focus

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cue / Progression
1	Barbell Bench Press	4×5-6	7→9	3 min	3-1-1	Scapula retracted, bar to lower sternum • +2.5 kg
2	Incline DB Press (30°)	3×8-10	8	2 min	2-0-1	Deep stretch • double progression
3	Seated DB Shoulder Press	3×8-10	8	90 s	2-0-1	Stop at ear level • add reps then load
4	Cable Fly / Pec Deck	3×12-15	9	60 s	2-1-1	Stretch + squeeze • add reps
5	Cable Lateral Raise	3×12-15	9	60 s	1-0-2	Lead with elbow, no shrug
6	Overhead Cable Triceps Ext	3×10-12	9	60 s	2-1-1	Long-head stretch overhead
7	Triceps Rope Pushdown	3×12-15	9	45 s	2-0-1	Elbows fixed, spread at bottom

Pull A — Back Thickness + Biceps

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cue / Progression
1	Weighted Pull-up / Lat Pulldown	4×6-8	7→9	2.5 min	2-0-1	Lead with elbows, full stretch • add reps then load
2	Pendlay / Chest-Supported Row	4×8-10	8	2 min	2-1-1	Pause at contraction • double progression
3	Seated Cable Row	3×10-12	8	90 s	2-1-2	Drive elbows back, squeeze
4	Face Pull / Reverse Pec Deck	3×15-20	9	45 s	1-1-1	Pinkies up (rear delt)
5	EZ-Bar Curl	3×8-10	9	75 s	2-0-1	No swing • double progression
6	Incline DB Curl	3×10-12	9	60 s	3-0-1	Full stretch behind body, supinate
7	Hammer Curl	2×12-15	9	45 s	2-0-1	Neutral grip (brachialis/forearm)

Legs A — Quad Focus

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cue / Progression
1	Back Squat	4×5-6	7→9	3 min	3-1-1	Brace 360°, hit depth • +2.5-5 kg
2	Leg Press	3×10-12	8	2 min	2-0-1	Control to ~90°, no slam
3	Romanian Deadlift	3×8-10	8	2 min	3-1-1	Push hips back, bar close
4	Leg Extension	3×12-15	10	60 s	2-1-1	1s squeeze at top
5	Seated Leg Curl	3×10-12	9	75 s	2-1-2	Slow eccentric
6	Standing Calf Raise	4×8-12	9	60 s	2-2-1	Full stretch + pause
7	Hanging Leg Raise	3×10-15	9	60 s	2-0-1	Posterior pelvic tilt

Push B — Shoulder Focus

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cue / Progression
1	Overhead Press (barbell)	4×6-8	7→9	2.5 min	2-0-1	Brace, bar over mid-foot • +2.5 kg
2	Incline Barbell/DB Press	3×8-10	8	2 min	2-1-1	Upper-chest squeeze
3	Machine Shoulder Press	3×10-12	9	90 s	2-0-1	Smooth path, no lockout rest
4	DB Lateral Raise	4×12-20	10	45 s	1-0-2	Slow lower, constant tension
5	Cable Fly	3×12-15	9	60 s	2-1-1	Deep stretch, hug
6	Close-Grip Bench / Dips	3×8-10	8	90 s	2-0-1	Elbows tucked
7	Triceps Pushdown	3×12-15	9	45 s	2-0-1	Full extension

Pull B — Lat Width + Biceps

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cue / Progression
1	Wide-Grip Lat Pulldown	4×8-10	8	2 min	2-1-2	"Elbows to floor", lat stretch up top
2	Chest-Supported Row	4×10-12	8	90 s	2-1-1	Full stretch + squeeze
3	Single-Arm DB Row	3×10-12	9	75 s	2-0-1	Big ROM, no torso twist
4	Straight-Arm Pulldown	3×12-15	9	45 s	2-1-1	Lat isolation, arms long
5	Reverse Pec Deck	3×15-20	9	45 s	1-1-1	Rear delt, pinkies lead
6	Cable Curl	3×10-12	9	60 s	2-0-1	Constant tension
7	Preacher / Spider Curl	2×12-15	10	45 s	2-1-1	No swing, peak squeeze

Legs B — Posterior / Glute-Ham Focus

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cue / Progression
1	Romanian Deadlift / Hip Thrust	4×8-10	8	2.5 min	3-1-1	Hinge / full glute lockout
2	Hack Squat / Pendulum	3×10-12	8	2 min	3-1-1	Deep ROM, quads loaded
3	Bulgarian Split Squat	3×10-12/leg	9	75 s	2-0-1	Slight torso lean, front-leg drive
4	Lying Leg Curl	3×12-15	9	60 s	2-1-2	No hip rise
5	Seated Leg Curl	3×12-15	9	60 s	2-1-2	Point toes, full squeeze
6	Seated Calf Raise	4×12-15	9	45 s	2-2-1	Soleus focus
7	Cable Crunch	3×12-15	9	45 s	2-1-2	Round spine, hips fixed

4 Weekly Volume vs. Evidence

Hard sets/week (direct + meaningful indirect) across the 6-day PPL/LPP. Anchor: Schoenfeld 2017 dose-response — 10+ sets/muscle/week drives growth; in a deficit, sit near the lower-effective end (MEV–MAV) to preserve muscle without exceeding recovery.

Muscle	Your Sets/wk	Evidence Target	Verdict
Chest	13-15	10-18	Optimal
Back (overall)	15-18	12-20	Optimal
Lats	9-11	8-14	Good
Upper back	6-8	6-12	Good
Front delts	9-11	6-12	Covered by pressing
Lateral delts	10-13	12-22	Solid
Rear delts	6-8	6-12	Good
Biceps	10-13	8-16	Optimal
Triceps	11-14	10-16	Optimal
Forearms	grip + 2-4	4-10	Adequate
Quads	11-13	10-18	Optimal
Hamstrings	9-11	8-16	Optimal
Glutes	8-10	6-16	Good
Calves	8-11	10-20	Good
Abs	6-9	6-16	Good

Read-out

Every muscle is trained 2×/week and sits at or above the 10-set growth threshold while staying within recoverable range for a dieting advanced lifter. Heavy compounds lead each session to protect strength — the #1 signal for keeping muscle in a deficit.

5 Progressive Overload (8 + 2 Weeks)

Core method = **Double Progression with RPE auto-regulation**. Add reps within a range until you top it on all sets, then add load (2.5 kg upper / 2.5–5 kg lower) and drop to the bottom.

Week	Phase	Intensity (RPE)	Volume	Action
1	Accumulation 1	7 (3 RIR)	Base	Establish loads, perfect technique
2	Accumulation 1	8 (2 RIR)	Base	Add reps across sets
3	Accumulation 1	8-9	+1 set lagging	Add reps; load up where maxed
4	Accumulation peak	9 (1 RIR)	Peak	Push compounds to top of range
5	DELOAD	5-6 (-40-50% load)	Half	Recover, movement quality, refeed
6	Intensification	8	Base	Reset heavier than W3, rebuild reps
7	Intensification	8-9	+1 set lagging	Add reps/load
8	Intensification peak	9 (1 RIR)	Peak	Top-set PRs where possible
9	Peak / Maintain	8-9	Slightly reduced	Hold strength, maximize definition
10	Peak / Finish	7-8 (taper)	Reduced	Maintain intensity, drop junk volume

In a deficit, success = **MAINTAINING** loads, not weekly PRs

Holding your numbers while losing 5–7 kg means you're getting relatively stronger and preserving muscle. That IS progress.

6 Nutrition

Maintenance Calories — Method Comparison

Method	BMR	×1.55	BF	LBM	Katch BMR
Mifflin-St Jeor	1,763	~2,730	20%	64.8 kg	1,770
Katch-McArdle (22% BF)	1,735	~2,690	22%	63.2 kg	1,735
			25%	60.8 kg	1,682

Maintenance ≈ 2,700 kcal/day

Activity ×1.55 (6 lifts + 10k steps + active life). Both methods agree → high confidence. Verify against 2-week weight trend.

Fat-Loss Calories

Approach	Deficit	Intake	Loss/wk	Muscle risk
Moderate	-15% (-400)	2,300	~0.4 kg	Very low (too slow)
Moderate-Aggressive	-22% (-600)	2,100	~0.6 kg	Low
Aggressive	-30% (-800)	1,900	~0.8 kg	Moderate-high

Chosen: ~2,100 kcal/day

Helms 2014 / Garthe 2011: ~0.5–0.7%/week BW loss maximizes fat loss while sparing muscle. ~0.6 kg/wk × 10 wks = ~6 kg → ~75 kg, landing in target while moderate enough (with high protein + heavy lifting) to protect muscle.

Macronutrients @ 2,100 kcal

Macro	Amount	g/kg	kcal	%	Rationale
Protein	190 g	2.35	760	36%	Helms: 2.3-3.1 g/kg LBM in deficit; preserves muscle + satiety
Fat	60 g	0.74	540	26%	Hormones, joints, EFAs
Carbs	200 g	2.5	800	38%	Training performance + recovery (keep most around training!)
Fiber	30-40 g	—	—	—	Satiety, gut health

Hydration & Electrolytes

Water	Sodium	Potassium	Magnesium
3-3.5 L/day	3,000-4,500 mg	3,500-4,700 mg	350-400 mg

Protein Distribution & Timing

- **5 meals**, every 3-4 h — ~0.4 g/kg/meal (Schoenfeld & Aragon 2018) maximizes daily MPS.
- **35-45 g/meal** (≥3 g leucine to cross the leucine threshold).
- **Pre-WO (AM):** ~25 g whey ± banana. **Post-WO:** ~40 g protein + carbs within 1-2 h.
- **Before bed:** ~30-40 g slow protein (paneer/Greek yogurt/milk) — Res 2012, overnight MPS.

7 Vegetarian Meal Plan

Sample Day — ~2,100 kcal | ~188 P / 58 F / 200 C

Meal / Time	Foods & Portions	P	F	C	kcal
Pre-WO 6:00	1 scoop whey + black coffee	24	2	2	120
Breakfast 8:30	50 g oats + 1 scoop whey + 150 g berries + 150 ml skim milk	40	7	55	430
Lunch 1:00	150 g tofu (or 120 g low-fat paneer) + 1 cup dal + 120 g rice + veg (1 tsp oil) + salad	33	16	65	540
Snack 4:30	200 g low-fat Greek yogurt + 15 g almonds	24	12	12	250
Dinner 8:00	55-60 g dry soya chunks (cooked) + 2 wheat rotis + large veg (1 tsp oil) + salad	39	11	56	480
Bed 10:00	100 g low-fat paneer or 200 g Greek yogurt	18	8	6	160
TOTAL	If short on protein, add ½ scoop whey (+12 P)	~188	~56	~196	~2,000-2,100

Meal Alternatives

Role	Options
Lean protein	Paneer, tofu, soya, tempeh, Greek yogurt, whey, skim milk
Legumes	Dal, rajma, chickpeas, beans, moong
Carbs	Oats, rice, roti, potato, sweet potato, fruit, quinoa
Fats	Almonds, walnuts, chia/flax, peanut butter, ghee/olive oil
Veg	Spinach, broccoli, capsicum, bhindi, cauliflower

Budget Swaps

Premium	Budget
Tofu/tempeh	Soya chunks (cheapest)
Greek yogurt	Hung curd
Whey isolate	Whey concentrate / milk+paneer
Almonds	Peanuts / roasted chana
Berries	Banana, papaya, seasonal fruit
Low-fat paneer	Homemade from toned milk

Use the companion Nutrition Tracker

Log your real meals in the interactive [nutrition-tracker.html](#) app to auto-sum calories, macros, fiber and key micros (calcium, iron, sodium, potassium, magnesium, B12, D) against these targets.

8 Supplements

Supplement	Status	Dose	Timing	Why
Creatine Monohydrate	ESSENTIAL	5 g/day	Any time, daily	Strongest evidence; vegetarians respond more
Whey Protein	ESSENTIAL	1-2 scoops	Pre/post WO	Convenient complete protein to hit 190 g
Vitamin B12	ESSENTIAL	250-500 mcg/day	With food	Not reliable in veg diet; nerves/blood/energy
Vitamin D3 (lichen)	ESSENTIAL	1,000-2,000 IU	With fatty meal	Common deficiency; bone/immune/performance
Omega-3 (Algae EPA/DHA)	RECOMMENDED	0.5-1 g EPA+DHA	With food	Veg diets low in EPA/DHA; recovery/joints/heart
Magnesium	OPTIONAL	200-400 mg	Evening	Often low in deficit; sleep/muscle
Electrolytes	OPTIONAL	As needed	Around training	High steps + sweat + low-sodium veg diet
Caffeine	OPTIONAL	3-6 mg/kg	30-45 min pre-WO	Performance, focus, appetite blunting
Citrulline Malate	OPTIONAL	6-8 g	Pre-WO	Modest pump/endurance
Beta-Alanine	OPTIONAL	3-5 g/day	Any time, daily	Helps 60-240 s high-rep efforts

Verdict

Essentials = Creatine, B12, D3, whey (practical), algae omega-3. Everything else is optional polish. No fat burners — they don't move the needle.

9 Cardio

Modality	Fat loss	Muscle/recovery cost	Verdict
Walking (NEAT)	High	Negligible	Keep 10k steps — your best tool
Incline treadmill (LISS)	High	Low	Primary structured cardio
Cycling (LISS)	Good	Low	Good alternative
HIIT	High/min	High	Max 1×/wk, away from leg days

- **Keep your 10,000 evening steps** — non-negotiable, your single best fat-loss tool. With 6 lifting days, do NOT add much structured cardio — let steps + the deficit do the work.
- **Add at most 1-2 LISS** sessions if fat loss stalls. Skip HIIT unless recovery is excellent (it competes with 6-day lifting).
- **Fasted cardio not superior** (Schoenfeld 2014, matched calories). Walking fasted is fine; for lifting take ~25 g whey first.

10 Recovery

Sleep (biggest recovery lever)

- Target 7.5-9 h; push toward 8+. Critical with 6 training days.
- Fixed times, dark/cool room (~18-20°C), no screens 60 min prior; cap caffeine early.

Warm-Up (RAMP, 8-10 min)

- 3-5 min easy cardio
- Dynamic mobility for the day's joints
- 2-3 ramp-up sets on first compound

Cool-Down

- 5 min easy walk + nasal breathing + light static stretch

Mobility (10 min, 3×/wk)

- Hip flexor, 90/90, deep squat hold, T-spine rotations, shoulder dislocates, hamstring/calf stretch

Foam Rolling

- Pre-training, tight areas 30-60 s each (warm-up adjunct)

Stress

- 10 min/day decompression — chronic stress + deficit = high cortisol, water retention masking fat loss

Recovering Well

Stable/rising lifts; good sleep; normal RHR; steady mood; normal appetite

Back Off / Deload

Strength dropping 2 sessions; RHR +5-10 bpm; poor sleep; persistent fatigue; joint aches; soreness >3 days

11 Tracking

Metric	How	Frequency
Body weight	Same time AM, fasted; record 7-day average	Daily → weekly avg
Waist	Tape at navel, relaxed, AM fasted	Weekly
Photos	Front/side/back, same lighting/poses	Weekly
Strength	Log load × reps × RPE (your true muscle-retention marker)	Every session
Body fat %	Calipers or visual vs. photos	Every 2-3 weeks
Nutrition	Log meals in the interactive tracker (macros + micros)	Daily

Calorie adjustment rule

Compare weekly averages. Losing 0.4-0.7 kg/wk → change nothing. >0.8 kg/wk for 2 wks → +150 kcal. <0.3 kg/wk for 2 wks → go to plateau tree. Waist + photos override the scale.

12 Plateau Strategy

Confirm a real stall first: weekly **average** weight flat for 2 weeks AND waist + photos unchanged. Water retention often masks fat loss for 1-2 weeks — don't panic early.

START Real stall confirmed (2 wks: weight + waist + photos all flat)

1 AUDIT FIRST — re-log food 3 days honestly (portion creep!). Hitting steps + sleep + protein? Fix adherence before changing the plan. *Most "plateaus" are this.*

2 ↑ **NEAT** — 10k → 12k steps or +1 LISS. Lowest cost to recovery; preserves muscle.

3 ↓ **Calories** modestly — cut ~150 kcal from carbs/fat (never protein). 2,100 → ~1,950.

4 + **Cardio** — add 1 LISS session (20-30 min). Keep HIIT ≤1×/wk.

5 **Check training** — if fatigued, REDUCE junk volume / use the 5-day fallback. Fatigue ≠ fat loss.

6 **Refeed / Diet break** — 1-2 days (or 7-14 days) at maintenance to restore leptin/performance & restart fat loss (Byrne 2018 MATADOR).

Re-evaluate after 2 weeks → repeat from top if needed.

Golden rules

Protein and lifting intensity are the LAST things you cut. Move NEAT/cardio and small calorie nudges first. A planned refeed/diet break is a tool, not a failure.

13 Expected Results (10 Weeks)

Metric	Start	Realistic Outcome	Notes
Body weight	81 kg	74-76 kg (-5 to -7)	~0.6 kg/wk
Body fat %	~22%	~14-16%	Visible abs, separation
Fat mass	~18 kg	~11-12 kg (-6 to -7)	Bulk of the loss
Lean body mass	~63 kg	maintain ± small gain	Preserve = the win
Waist	baseline	-5 to -8 cm	Strongest visual change
Strength	baseline	Maintained	Holding loads = success
Appearance	soft ~22%	Lean, athletic, vascular, defined	Dramatic visual transformation

The honest truth

As an advanced lifter in a deficit, large simultaneous muscle gain is unlikely (that's a beginner phenomenon). Your excellent, achievable outcome: lose ~6-7 kg fat, preserve essentially all muscle, possibly add a little. Far leaner → you'll LOOK much more muscular as hidden muscle becomes visible. 100% achievable naturally.

Daily

- ~2,100 kcal / ~190 g protein
- 5 protein-spread meals (incl. pre-bed slow protein)
- 3-3.5 L water + salt food
- 10,000 steps
- Creatine 5 g + B12 + D3 + algae omega-3
- 7.5-9 h sleep
- Log workout (load × reps × RPE)
- Log meals in the nutrition tracker

Weekly

- 6 sessions: Push/Pull/Legs ×2 (or 5-day fallback)
- Steps daily; LISS only if needed
- Weigh-in: 7-day average
- Waist + progress photos
- Review: strength holding? losing ~0.6 kg? adjust
- Mobility 3× + recovery check

Phase Milestones

- **Week 5:** Deload (light, recover, refeed)
- **Weeks 6-8:** Intensification — chase rep/load PRs where recovery allows
- **Weeks 9-10:** Peak/finish — maintain intensity, trim junk volume, lock in the lean look

Your complete protocol

Single, fully-optimized plan for your 174 cm / 81 kg advanced vegetarian profile, morning training, high step count, and 10-week lean-out goal — now on the 6-day PPL/LPP structure you prefer. Execute the levers in order, judge muscle retention by STRENGTH not the mirror, keep the heavy work heavy, and protect sleep + protein above all. 15% body fat at ~75 kg is well within reach.

This document provides general evidence-based fitness and nutrition information, not personalized medical advice. As a long-term vegetarian entering a sustained deficit, consider a baseline blood panel (vitamin D, B12, ferritin, lipids) and consult a qualified professional before making significant changes.